Group II, Row IV: extraction of the DGLAP, RC, BFKL and finite terms, starting from the expression after the JIMWLK subtraction

1 The starting expression and its check

After extracting only the JIMWLK term, the remainder is (the correction below is marked in red)

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^4}\,\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\bar\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\mathcal{P}_{gg}(\xi)-\frac{\beta_0}{2}\,\delta(1-\xi)\right]\\
&\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^4}\,\frac{1}{k^+}\,\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\right]\\
&\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{k^+/\vee}^1 d\xi\,\frac{1-\xi}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\bar\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\right]\\
&\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)\,. \qquad (1)
\end{aligned}
$$

The one correction. The prefactor of the first term was written as $\frac{g^4N_c}{8\pi^4}$; it must be $\frac{g^4}{16\pi^4}$, i.e. smaller by $2N_c$. The reason: the splitting function already carries its own colour factor,

$$
\mathcal{P}_{gg,+}(\xi)=2N_c\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{(1-\xi)_+}\Big]\,,\qquad \mathcal{P}_{gg}(\xi)=\mathcal{P}_{gg,+}(\xi)+\frac{\beta_0}{2}\,\delta(1-\xi)\,,\qquad \beta_0=\frac{11N_c}{3}\,, \qquad (2)
$$

while the regrouped expression that this descends from reads $-C\int\frac{d\xi}{\xi^2}\,\hat p(\xi)\,L_\epsilon\,(\dots)$ with $C=\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}$ and the colour-*free* kernel $\hat p(\xi)=\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}$. Substituting $\hat p_+=\frac{1}{2N_c}\mathcal{P}_{gg,+}$ requires dividing the prefactor by $2N_c$: $\frac{C}{2N_c}=\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^4}\frac{1}{k^+}$. Independent cross-check: the $-\frac{\beta_0}{2}\delta(1-\xi)$ piece must reproduce the running-coupling term $+\frac{g^2\beta_0}{16\pi^2}L_\epsilon\frac{d^3N_{\rm LO}}{d^3k}$ (Step 1 below); with $\frac{g^4}{16\pi^4}$ it does, with $\frac{g^4N_c}{8\pi^4}$ it would come out $2N_c$ times too large. The second and third terms of (1) are correct as written.

Notation. $C\equiv\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}$, $\bar\xi=\frac{k^+}{\vee}$, $L_\epsilon=\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}$,

$$
W(u,u')\equiv\Big[U^{ce'}(u')-U^{ce'}(x')\Big]\,[U^{ce}(u)-U^{ce}(x)]\,\rho^{e'}(x')\,\rho^e(x)\,, \qquad (3)
$$

and the ξ-dependent LO structure

$$
\frac{d^3N_{\xi\,\rm LO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\bar\xi}}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y') \qquad (4)
$$

($\xi=1$ is the LO cross-section). With these, the (corrected) first term of (1) is

$$
A=-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\Big[\mathcal{P}_{gg}(\xi)-\frac{\beta_0}{2}\delta(1-\xi)\Big]\,L_\epsilon\,\frac{d^3N_{\xi\,\rm LO}}{d^3k}\,, \qquad (5)
$$

since $\frac{g^2}{8\pi^2}\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}=\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^4}\frac{1}{k^+}$. Note that $\mathcal{P}_{gg}-\frac{\beta_0}{2}\delta(1-\xi)=\mathcal{P}_{gg,+}$ by (2), and the plus prescription in (5) is the *locally subtracted* one on $[\bar\xi,1]$, $\int_{\bar\xi}^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi}$ — that this is the convention in force is visible in (1) itself, because the endpoint term with $\log(1-\frac{k^+}{\vee})$ (second term) appears *explicitly*; in the standard $[0,1]$ convention it would be absorbed into the plus distribution. The second term of (1) is $-C\,L_\epsilon^{\rm LO}\log(1-\bar\xi)$ on the LO structure (the leftover of the endpoint logarithm after the log $\frac{k^+}{\Lambda}$ part left with JIMWLK), and the third term is the azimuthal term (its two prefactors combine to the same C, since $\frac{g^2}{4\pi^2}\cdot\frac{g^2}{2\pi^2}N_c=\frac{g^4N_c}{8\pi^4}$).

2 Step 1: the DGLAP and RC terms

Split the bracket of (5) into its two pieces, $\mathcal{P}_{gg}$ and $-\frac{\beta_0}{2}\delta(1-\xi)$:

$$A = \underbrace{-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}}_{\text{DGLAP}} + \frac{g^2}{8\pi^2}\,\frac{\beta_0}{2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,\delta(1-\xi)\,L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\,. \tag{6}$$

The δ-integral is elementary: at $\xi=1$ the measure factor is $\frac{1}{\xi^2}=1$ and the phase of (4) collapses to the LO phase, so $\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\big|_{\xi=1} = \frac{d^3N_{\mathrm{LO}}}{d^3k}$, and

$$A = \underbrace{-\frac{g^2}{8\pi^2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,\mathcal{P}_{gg}(\xi)\,L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}}_{\text{DGLAP}} + \underbrace{\frac{g^2\,\beta_0}{16\pi^2}\,L_\epsilon\,\frac{d^3N_{\mathrm{LO}}}{d^3k}}_{\text{RC}}\,. \tag{7}$$

The RC term carries the pure-glue $\beta_0 = \frac{11N_c}{3}$ and enters the coupling renormalization; the DGLAP term is the physical collinear kernel of the measured gluon. The DGLAP term, however, still contains the soft channel $2N_c\frac{1-\xi}{\xi}$ of $\mathcal{P}_{gg}$, with a hidden rapidity logarithm at $\xi\to 0$ multiplying L_ϵ. The next two steps deal with it.

3 Step 2: constructing the subtraction term, explicitly

(2a) Two building blocks. The first is (4). The second is the dimensionally regularized transverse integral derived in the companion reduction note,

$$\int_z^{(d_\perp=2-\epsilon)} K(y,y',z) = -\pi\,L_\epsilon\,, \qquad K(a,b,z) = \frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}\,, \tag{8}$$

which we read *backwards*:

$$L_\epsilon = -\frac{1}{\pi}\int_z K(y,y',z)\,. \tag{9}$$

Eq. (9) trades the abstract ultraviolet bracket L_ϵ for a concrete transverse integral over one new point z. That is the whole trick of Step 2: once L_ϵ is a z-integral, a counterterm can be built under the *same* integral sign and compared with it point by point.

(2b) The term that needs a partner. The DGLAP term of (7) contains the soft channel

$$D_{\mathrm{soft}} \;=\; -\frac{g^2}{8\pi^2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}\,. \tag{10}$$

It has the two dangerous features at once: the ultraviolet bracket L_ϵ, and a hidden $\log\frac{\vee}{k^+}$ from its $\xi\to 0$ endpoint ($\xi\to 0$ is $p^+\to\vee$: the unmeasured gluon carries almost all of the projectile momentum, the measured gluon is the soft one — the projectile-side rapidity region). A cross-section must not contain $\log\frac{\vee}{k^+}\times L_\epsilon$. The subtraction term is built to reproduce *exactly* the $\xi\to 0$ behaviour of (10) — nothing more — so that in the difference the rapidity logarithm and the μ-dependence disappear together.

(2c) Put D_{soft} under a z-integral. Insert (9) into (10) and write out (4):

$$D_{\mathrm{soft}} = +\frac{g^2}{8\pi^2}\int_{\bar\xi}^1 \frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{1}{\pi}\,K(y,y',z)\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')\,. \tag{11}$$

This is still exact; the overall sign flipped because of the minus sign in (9).

(2d) Fast and slow factors at $\xi\to 0$. The phase ties $y'-y$ to $O(\xi)$; concretely, in the variables $y = z+\xi(w-z)$, $y' = z+\xi(w'-z)$ (the exact map that Step 3 uses), $\xi\to 0$ at fixed (w,w',z) sends both parent points onto the point z: $y\to z$, $y'\to z$. The limit treats the factors of (11) very differently:

fast: $K(y,y',z) = \dfrac{K(w,w',z)}{\xi^2}$ blows up — this is what generates the logarithm;

slow: $\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') \;\to\; \mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) = K(x,x',z)$ and $W(y,y')\to W(z,z)$ — smooth, finite limits.

(2e) Define the counterterm by freezing the slow factors. The subtraction term is obtained from (11) by *one* rule: keep the phase and the fast factor untouched, and replace the slow factors by their collapse-point values,

$$\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y') \;\longrightarrow\; K(x,x',z)\,W(z,z)\,, \tag{12}$$

with an overall minus sign so that it subtracts. Explicitly, per ξ,

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\pi}\,K(y,y',z)\,K(x,x',z)\,W(z,z)\,, \tag{13}$$

and, summed over ξ with the same measure, kernel weight and prefactor as (10),

$$S \;\equiv\; -\,\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{d^3N_\xi^{\rm sub}}{d^3k} \;=\; -\,\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,\frac{1-\xi}{\xi}\int_{x,y,x',y',z} e^{-ik(y'-y)\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\,\Big[U^{ce'}(z)- \tag{14}$$

Nothing new has been invented: the $\frac{1}{\pi}$, the prefactors and the measure of (13)–(14) are all inherited from (11). Physically, (13) is the real-emission structure with the parent at the common point z — the two soft-emission (dipole) kernels $K(y,y',z)\,K(x,x',z)$ of amplitude and conjugate, and the Wilson lines at coincident points — precisely the building blocks of the real BFKL kernel. It is IR and UV finite in $d=2$ and contains no μ.

(2f) What the pair achieves. Adding (11) and (14), the two integrands sit under the same integral sign and combine into a single bracket:

$$D_{\rm soft}+S = +\,\frac{g^2}{8\pi^2}\int_{\bar\xi}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{K(y,y',z)}{\pi}\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')-K(x,x',z)\,W \tag{15}$$

By (2d) the square bracket vanishes *linearly* in ξ at fixed (w,w',z), and in those variables the z-integral converges at large z (the parent points travel with z, so both terms of the bracket fall off together); so the pair is free of $\frac{2}{\epsilon}$, γ_E and μ, and its soft region contributes $\int\frac{d\xi}{\xi}\times O(\xi)=$ finite — no $\log\frac{\vee}{k^+}$. The subtraction S alone therefore carries all the ultraviolet data and the entire rapidity logarithm of $D_{\rm soft}$. And S, unlike $D_{\rm soft}$, can be computed in closed form: that is Step 3.

4 Step 3: evaluating S in closed form, explicitly

(3a) Only the phase knows about ξ. Look at the inner integral of (14) at fixed ξ, with the ξ-independent factors pulled out of the y,y' integration:

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,x',z} K(x,x',z)\,W(z,z)\,\frac{1}{\pi}\int_{y,y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,K(y,y',z)\,. \tag{16}$$

The variable ξ appears in exactly one place: the phase. Strategy: change variables in the y,y' integral so that the phase loses its $\frac{1}{\xi}$; then every power of ξ is explicit, and the ξ-integral factorizes from the transverse one.

(3b) The substitution. At fixed z set

$$y = z+\xi\,(w-z) \iff w = z+\frac{y-z}{\xi}\,,\qquad y' = z+\xi\,(w'-z) \iff w' = z+\frac{y'-z}{\xi}\,, \tag{17}$$

an exact, invertible, linear change of variables — there is no approximation anywhere in Step 3.

(3c) Transform each factor, one at a time.

(i) Measure (two transverse dimensions each): $d^2y=\xi^2\,d^2w$, $d^2y'=\xi^2\,d^2w'$, hence $d^2y\,d^2y'=\xi^4\,d^2w\,d^2w'$.

(ii) Differences: $y-z=\xi\,(w-z)$, $y'-z=\xi\,(w'-z)$, and, subtracting the first from the second, $y'-y=\xi\,(w'-w)$.

(iii) Kernel:

$$K(y,y',z) = \frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2} = \frac{\xi^2\,(w-z)\cdot(w'-z)}{\xi^2(w-z)^2\;\xi^2(w'-z)^2} = \frac{1}{\xi^2}\,K(w,w',z)\,. \tag{18}$$

(iv) Phase:

$$e^{-ik\cdot(y'-y)\frac{1}{\xi}} = e^{-ik\cdot\xi(w'-w)\frac{1}{\xi}} = e^{-ik\cdot(w'-w)}\,. \tag{19}$$

The $\frac{1}{\xi}$ is eaten *exactly* — nothing is expanded or approximated.

(v) Spectators: $K(x,x',z)$, $W(z,z)$ and the x,x',z measure contain no y,y' and are untouched.

(3d) Assemble. Putting (i)–(v) into (16),

$$\frac{d^3N_\xi^{\rm sub}}{d^3k} = \xi^4\cdot\frac{1}{\xi^2}\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\frac{1}{\pi}\,\mathcal{I} = \xi^2\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\frac{1}{\pi}\,\mathcal{I}\,,\qquad \mathcal{I}\equiv\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,W(z,z)\,, \tag{20}$$

and $\mathcal{I}$ is manifestly ξ-independent: *all* the ξ-dependence of $\frac{d^3N_\xi^{\rm sub}}{d^3k}$ is the explicit factor ξ^2 (Jacobian ξ^4 times kernel rescaling ξ^{-2}).

(3e) The ξ-integral and the prefactor. Insert (20) into (14): the explicit $\frac{1}{\xi^2}$ of the measure is cancelled by the ξ^2 of (20), leaving

$$\int_{\bar\xi}^1 d\xi\,\frac{1-\xi}{\xi} = \int_{\bar\xi}^1\frac{d\xi}{\xi} - \int_{\bar\xi}^1 d\xi = \Big[\log\xi\Big]_{\bar\xi}^1 - \Big[\xi\Big]_{\bar\xi}^1 = \big(0-\log\bar\xi\big) - \big(1-\bar\xi\big) = \log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big) \tag{21}$$

(upper minus lower in both brackets; positive, as required for a positive integrand), and the prefactor assembles as

$$\frac{g^2}{8\pi^2}\cdot 2N_c\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\cdot\frac{1}{\pi} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\,. \tag{22}$$

Therefore

$$S = -\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x) \tag{23}$$

5 Step 4: add and subtract, and the BFKL term

Insert $0 = S - S$ into the decomposition, keeping $+S$ in its integral form (14) (it goes to the finite block, where it protects the DGLAP soft channel via (15)) and replacing $-S$ by minus the closed form (23). Since (23) carries an overall minus sign, $-S$ is *plus* $\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^5}\frac{1}{k^+}\big[\log\frac{\vee}{k^+} - (1-\frac{k^+}{\vee})\big]\,\mathcal{I}$. Its $\log\frac{\vee}{k^+}$ part is the **BFKL contribution of this row**,

$$\boxed{\frac{d^3N^{\rm BFKL}_{\text{G-II, R IV}}}{d^3k} = +\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\log\Big(\frac{\vee}{k^+}\Big)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\,\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)} \tag{24}$$

moved to the BFKL file (it appears there verbatim); its coefficient is manifestly UV finite and μ-independent, as a BFKL kernel must be. The $-(1-\frac{k^+}{\vee})$ part of $-S$ is the boundary term of the finite block.

6 Result

Collecting Steps 1–4, the starting expression (1) decomposes as

$$\frac{d^3N_{\rm NLO}}{d^3k} = \text{DGLAP} + \text{RC} + \frac{d^3N^{\rm BFKL}}{d^3k}\Big|_{(24)} + \frac{d^3N^{\rm finite}}{d^3k}\,, \tag{25}$$

with DGLAP and RC given in (7) and

$$\frac{d^3N^{\text{finite}}}{d^3k} = \underbrace{-\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\;L_\epsilon\;\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\,[U^{ce}(y)-U^{ce}(x)]\,/}_{\text{endpoint (plus-prescription convention) term}}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\frac{(1-\xi)}{\xi}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,K(y,y',z)\,K(x,x',z)\;\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x}_{\text{soft subtraction } (=\mathrm{S})}$$

$$\underbrace{-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,K(w,w',z)\,K(x,x',z)\;\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,[U^{ce}(z)-U^{ce}(x)]\,\rho^{e'}(x'}_{\text{boundary}}$$

$$\underbrace{+\frac{1}{(2\pi)^3}\frac{g^4N_c}{8\pi^4}\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]}_{\text{azimuthal}} \tag{26}$$

Bookkeeping

(i) (soft subtraction S) + (boundary) + (BFKL entry (24)) = 0 identically: Steps 2–4 are an exact add-and-subtract of (23), and all three objects are UV finite. Together with Step 1 being an exact split, the decomposition reproduces the (corrected) starting expression (1) identically.

(ii) The role of S in the finite block is to pair with the soft channel $2N_c\frac{1-\xi}{\xi}\otimes L_\epsilon$ of the DGLAP term: per ξ, their sum is the μ-free combination (15), whose bracket is $O(\xi)$, so the pair contains no $\log\frac{\vee}{k^+}\times L_\epsilon$.

(iii) The endpoint term (first entry of (26)) is the difference between the two plus-prescription conventions. If $\mathcal{P}_{gg}$ is used with the standard $[0,1]$ plus distribution, this entry is absorbed into the DGLAP term and disappears from (26); nothing else changes.

(iv) Neither the endpoint term nor the azimuthal term contains a rapidity logarithm: $\log(1-\frac{k^+}{\vee})\to 0$ as $\vee\to\infty$, and the azimuthal ξ-integral converges at $\xi\to 0$ (oscillating phase; the collapse region is suppressed by the measure and by the exactly vanishing angular average of the azimuthal bracket). The only rapidity logarithm of this expression is the one carried by the DGLAP soft channel and rerouted into (24).